Vedant Dobwal

Undergraduate Researcher — Theoretical Physics and Mathematics
Email: vedant24@iiserb.ac.in
Research Website: wristyspider834.github.io/vedantdobwal
Google Scholar: scholar.google.com/citations?user=wFNvX78AAAAJ
LinkedIn: linkedin.com/in/vedant-dobwal-738234207
ResearchGate: researchgate.net/profile/Vedant-Dobwal
ORCID: 0009-0005-1124-0944

Education

Indian Institute of Science Education and Research (IISER) Bhopal Ongoing
BS–MS (Natural Sciences)
Intended Major: Mathematics

Indian Institute of Technology (IIT) Guwahati Concurrent
Certificate Program — Data Science & Artificial Intelligence

Research Experience

Theoretical Research — Plasma Physics & Nonlinear Dynamics (2024–Present)
Focus: Maxwell-compatible dynamical frameworks, controlled chaos, emergent geometry, and magnetohydrodynamic (MHD) theory.

- Developed a theoretical model investigating the role of guided chaotic behavior in stabilizing plasma confinement domains.
- Derived perturbative extensions consistent with Maxwell's equations and standard conservation principles.
- Analyzed transitions between diffusive and localized attractor regimes and examined their implications for confinement dynamics.

Derivation of Controlled-Chaos Framework in Maxwell–MHD Systems

- Derived stabilization-like terms from the electron momentum equation and generalized Ohm's law.
- Clarified physical interpretation of perturbative control fields and their influence on stability.
- Produced technical notes to support transparency and reproducibility of derivations.

Fluid–Structure Dynamics & Vestibular Modelling (2025)
Focus: Contraction-based analysis, symmetry breaking, and reduced fluid–structure models.

- Re-analysed a recent reduced semicircular-canal model (J. Fluid Mech., 2025) from a dynamical-systems perspective using a scalar specialization of a general contraction criterion.

- Derived global exponential contraction of the reduced cupular pressure dynamics with an explicit rate, and obtained small-/large-stiffness asymptotics explaining velocity- vs. acceleration-tracking regimes.
- Identified an algebraic stiffness threshold at which asymmetric flow corrections dominate the symmetric axial velocity, providing a quantitative condition for symmetry breaking without loss of well-posedness.

Neutral Geometry and Necessary Conditions for Chaos in Dynamical Systems
Preprint in preparation 2025–2026

- Developed a geometric constraint framework for smooth autonomous dynamical systems based on instantaneous metric deformation induced by the symmetric Jacobian.
- Derived necessity conditions for invariant sets, recurrence, bifurcations, and chaotic dynamics through a coordinate-invariant partition into contractive, stretch, and neutral regions.
- Obtained dimension bounds on invariant sets and introduced a planar reduction yielding a scalar discriminant formulation that recovers classical bifurcations without explicit spectral analysis.

On a Circulation Functional Associated with Planar Vector Fields
Preprint in preparation 2025–2026

- Introduced a scalar circulation functional based on the symmetric part of the Jacobian to measure accumulated tangential metric deformation along closed curves in planar dynamical systems.
- Derived structural relations linking zeros and higher-order vanishing of the functional to the existence, stability, and multiplicity of limit cycles via a geometric degeneracy index.
- Obtained degree-dependent bounds for polynomial vector fields and demonstrated consistency with classical settings including Hopf-type systems, Liénard systems, and near-Hamiltonian perturbations.

Conferences & Highlights

- Feature — MIT OpenCourseWare (MIT OCW), News & Media, Massachusetts Institute of Technology (2026): ocw.mit.edu.
- NUPC Conference — Research Presentation (2025) : https://www.linkedin.com/in.

Conference Presentations

European Physical Society (EPS 52nd Plasma Physics Conference 2026)
Poster Presentation (Accepted) 2026

- *Controlled chaos framework for localized plasma confinement in Maxwell-compatible magnetic structures.*
- Presented a theoretical framework exploring guided chaotic dynamics for stabilization of confined plasma domains within Maxwell-consistent magnetic field configurations.

52nd IOP Annual Plasma Physics Conference
Poster Presentation (Accepted) 2026

- *Dynamic Control of Chaotic Attractors in Maxwell-Consistent Field-Constrained Plasmas.*
- Investigated mechanisms for controlling chaotic attractors in plasma systems subject to Maxwell-consistent field constraints and nonlinear dynamical feedback.

Skills

Mathematics: Dynamical systems, stability analysis, contraction theory, bifurcation analysis, multivariable calculus.
Physics: Magnetohydrodynamics, plasma modeling, electrodynamics.
Computational: Python (NumPy, Matplotlib), numerical experimentation, symbolic derivations.
Research: LaTeX, scientific writing, independent literature analysis.

Research Interests

Nonlinear dynamical systems, geometric methods in dynamical systems, chaos and bifurcation theory, plasma physics and magnetohydrodynamics (MHD), hydrodynamic instabilities, pattern formation in reaction–diffusion systems, and mathematical structures underlying complex physical systems.

Affiliations & Activities

- IISER Bhopal — Undergraduate Researcher
- American Physical Society — Student Member
- NASA Citizen Science Volunteer
- Coursework and academic enrichment through CERN, University of Pennsylvania, UCL/UK learning programs